

Adriano Machado

☎ +351 919 213 129 | ✉ adriano.alexandre.machado1@gmail.com | 🌐 adrianomachado.pt | 📺 Adriano-7 | 🌐 [adriano-alexandre-machado](https://www.linkedin.com/in/adriano-alexandre-machado)

Education

University of Porto

M.Sc. in Artificial Intelligence [🔗](#)

Porto, Portugal

Sep. 2024 – Jul. 2026

- Final grade: 17/20
- Thesis: Evaluated the performance of open-weight LLMs in negotiation scenarios, evaluating the impact of Personas, Self-Refine and multi-agent deliberation. (Grade 18/20)
- Relevant Coursework: Natural Language Processing, Computer Vision, Deep & Reinforcement Learning.

University of Porto

B.Sc. in Informatics and Computer Engineering [🔗](#)

Porto, Portugal

Sep. 2021 – Jul. 2024

- Final grade: 15/20
- Relevant Coursework: Algorithms & Data Structures, Operating Systems, Software Design & Testing.

Escola Secundária da Lixa

High School, Science and Technology

Lixa, Portugal

Sep. 2018 – Jun. 2021

- Final grade: 19/20

Work Experience

Faculty of Engineering of the University of Porto

Student Researcher

Porto, Portugal

Jul. 2025 – Oct. 2025

- Investigated how class imbalance (up to 100:1) in training data affects fully synthetic tabular data generation.
- Benchmarked 5 generators (**CTGAN**, **TVAE**, **RTVAE**, **DDPM** and **SMOTE**) on 6 datasets.
- Showed that downstream minority-class F1 degrades under severe imbalance, most for CTGAN and DDPM. The generators also increasingly failed to produce the balanced output requested, and standard aggregate quality metrics (Density, Coverage, Authenticity) hid minority-class degradation.
- Worked with **Python**, **Scikit-Learn**, **Synthcity**, and **pyMDMA**.

Jumpseller

Intern

Porto, Portugal

Feb. 2024 – Jun. 2024

- Worked on the email automations feature of Jumpseller's platform (**React/Redux**).
- Built 12 dynamic, multilingual email templates using live store data (**Ruby on Rails, Liquid**).
- Redesigned the template selection interface, adding real-time email previews (**React, Redux**).

Student Associations

Erasmus Student Network (ESN) Porto

Volunteer & IT Manager

Porto, Portugal

Feb. 2025 – Aug. 2026

- Led the IT department of the Porto section of Europe's largest student association, which welcomes 3500+ exchange students each semester.
- Won the **Lobos d'Ouro** award for **Best National IT Initiative**.
- Built an internal asset requisition platform and a scholarship evaluation platform used to review hundreds of applications; built the section's new public website.
- Automated the generation of event descriptions for the public website and built a predictive model to forecast ESN Card demand.

SINF (Semana de Informática)

Head of the Program Department

Porto, Portugal

Jun. 2025 – Oct. 2025

- Led the department responsible for the program: 11 talks and 8 workshops, the **largest program in six years**, up from 4 talks and 2 workshops the previous edition.
- Hosted **Steven Pemberton**, a researcher at CWI Amsterdam and co-author of HTML and CSS, as SINF's first major international speaker.
- Coordinated speaker outreach, scheduling, and on-site logistics with a team of volunteers.

ENEI (National Meeting of Informatics Students)

Member of the Program Department

Porto, Portugal

Sep. 2024 – Apr. 2025

- Recruited speakers and curated the program for a national student tech conference: 25 talks and 21 workshops.
- Handled outreach, scheduling, and speaker logistics.

NIAEFEUP (Informatics Students' Association)

Member of UNI's development team

Porto, Portugal

Oct. 2023 – Oct. 2025

- Member of the 3-person UI/UX team that led the redesign of **UNI**, an open-source mobile app used by University of Porto students.
- Member of the development team implementing the redesign and contributing to ongoing features.
- Worked with **Flutter**.

Projects

More projects and detailed write-ups are available at adrianomachado.pt.

Open-Weight LLMs as Negotiators (MSc Thesis)

[More Details](#)

LLMs · Hugging Face Transformers, SLURM

2026

- Benchmarked 9 open-weight LLMs across Gemma, Mistral, and Qwen families (4B–27B) in 3 game-theoretic negotiation scenarios (Ultimatum, Trading, BuySell), extending NegotiationArena.
- Evaluated **Self-Refine** and **team deliberation**; Qwen had the best performance (63–67% win rate) and Self-Refine improved the performance of disadvantaged seats by 5.6 percentage points.

Politeness Classification, from SVMs to LoRA

[More Details](#)

NLP · PyTorch, Hugging Face Transformers, W&B

2025

- Four-way politeness classifier on Intel’s Polite Guard dataset, from classical ML baselines to fine-tuned transformers and parameter-efficient adapters.
- Fine-tuned **RoBERTa** with domain adaptation achieved 0.924 F1, matching the published state of the art.
- LoRA** adapters saved ~25% training time with minimal performance loss (0.901 F1), outperforming **IA3** across all tested models.

Chessboard Photo to FEN

[More Details](#)

Computer Vision · OpenCV, Ultralytics YOLO, PyTorch, Optuna

2025

- Built a digital twin pipeline that converts physical chessboard photographs into FEN notation.
- Hybrid pipeline (**YOLOv8x** + **U-Net** corner heatmaps) achieved 0.02 edit distance per board.
- YOLOv8x piece detection reached 0.995 mAP50.

Skills

Programming Languages	Python, Java, C, C++, JavaScript, Dart
AI & Data Science	PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
Frameworks & Libraries	Flutter, React, Ruby on Rails
Tools & Platforms	Git, Docker, Linux, Firebase, Figma, SLURM, vLLM
Languages	Portuguese (native), English (fluent)